Homework Assignment 1

William Jarkander, Erik Selderman, Fabian Lindh, Jennifer Rundkvist

January 2026

Task 1

a)

We have the model:

$$y_i = e^{\beta_0} e^{\beta_1 x_i} x_i^{\beta_2} e^{\epsilon_i}$$

The task is to find the least square estimate of $\boldsymbol{\beta} = [\beta_0, \beta_1, \beta_2]^T$. Since the model is nonlinear, to find the least squares estimate, we first have to take the log of the function:

$$\log(y_i) = \log(e^{\beta_0} e^{\beta_1 x_i} x_i^{\beta_2} e^{\epsilon_i})$$

$$\log(y_i) = \beta_0 + \beta_1 x_i + \beta_2 \log(x_i) + \epsilon$$

After the log transformation, we use the following theorem to find the least squares estimator:

$$\hat{\boldsymbol{\beta}} \mid \mathbf{X} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}. \tag{1}$$

After some calculations in R, this results in:

$$\hat{\boldsymbol{\beta}} = [\hat{\beta}_0 \approx 5.362, \hat{\beta}_1 \approx -0.449, \hat{\beta}_2 \approx 1.114]^T$$

b)

The next task is to create a scatter plot of the data and a fitted regression model $e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_i} x_i^{\hat{\beta}_2}$. This results in the following plot:

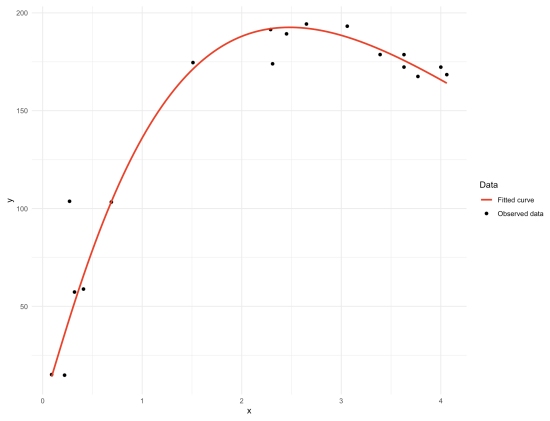


Figure 1: Fitted regression model against x

c)

Now we have to analytically find the value x that maximizes:

$$\hat{y}_i = e^{\hat{\beta}_0} e^{\hat{\beta}_1 x_i} x_i^{\hat{\beta}_2}$$

Begin by taking the log for easier calculations:

$$\log(y_i) = \beta_0 + \beta_1 x_i + \beta_2 \log(x_i) + \epsilon$$

We take the derivative with respect to x and set to 0:

$$\hat{\beta}_1 + \frac{\hat{\beta}_2}{x} = 0$$

$$x = -\frac{\hat{\beta}_2}{\hat{\beta}_1}$$

Calculating this value in R results in:

$$x = 2.481163$$

We also check if this is the maximum point by taking the second derivative:

$$-\frac{\hat{\beta}_2}{x^2} \approx -0.18 < 0$$

We see that this is a maximum since it is negative

Task 2

The next task is to see the effect of omitted variable bias with a Monte Carlo simulation. The model is given

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon. \tag{2}$$

Where $\varepsilon \sim \boldsymbol{N_n}(\mathbf{0}, \boldsymbol{\sigma^2 I})$, $\boldsymbol{\beta} = [20, 0.03, 0.7, 1.4]$ and $\sigma^2 = 6$. The simulation is designed to omit the variable x_2 from the model and then estimate the parameters in the reduced model to and then run the simulation $s = 300$ times and then compare the simulated parameters to the true values.

Before running the simulation we break down some theoretical expectations. We know if we omit x_2 and estimate the misspecified model:

$$y = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \tilde{\beta}_3 x_3 + u,$$

then the expected value of the OLS estimator $\tilde{\beta}_1$ is:

$$E[\tilde{\beta}_1 \mid X] = \beta_1 + \gamma_1 \beta_2 \tag{3}$$

where γ_1 is the coefficient of x_1 in the auxiliary regression:

$$x_2 = \gamma_0 + \gamma_1 x_1 + \gamma_3 x_3 + v. \tag{4}$$

We know $\beta_2 = 0.7$ and the estimated coefficient for γ_1 is around 0.0004, this make us conclude that β_1 from the misspecfied model will overestimate the true effect.

Now going to the simulation, y is generated by the Equation 2, then while omitting x_2 from the model, $\boldsymbol{\beta}$ is estimated by

$$\hat{\boldsymbol{\beta}} \mid \mathbf{X} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}. \tag{5}$$

The parameters from the misspecified model are estimated 300 times and then saved. We begin with study the results of β_0^s & β_3^s before doing a deeper investigation into β_1^s

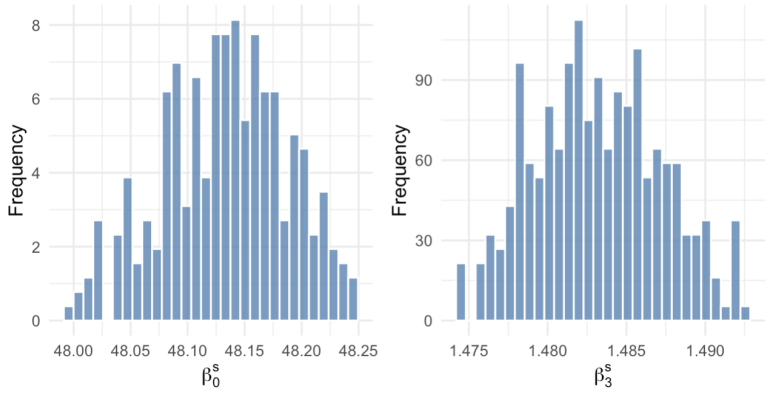


Figure 2: Histogram of β_0^s & β_3^s Misspecifed model

As we can see in Figure 2 the true value of $\beta_0 = 20$ and simulated β_0^s seem to differ significantly, compared to the true value of $\beta_3 = 1.4$ and the simulated β_3^s the difference does not seem to be as big, however the density of β_3^s does not seem to cover the true value of β_3.

Moving onward to the effect of β_1 when omitting the variable x_2 a deeper investigation is done.

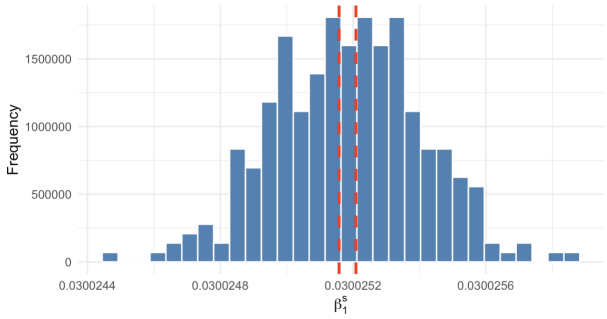


Figure 3: Histogram of β_1^s Misspecifed model

As can be seen Figure 3 the sampling distribution of β_1^s and the two dashed red lines represent the 95% confidence interval. The simulated β_1^s is slightly above the true value which aligns with the expectations from Equation 3

	Mean	CI Lower	CI Upper
β_1^s	0.03002518	0.03002516	0.03002521

Table 1: Caption

In Table 1 the mean of β_1^s and the confidence interval of the sampling distribution is presented.

The bias from the simulation from the misspecified model is around 0.000025 which aligns with the expectations of overestimating the true value. The bias is relative small, we can see that the 95% confidence interval does not cover the true value of $\beta_1 = 0.03$.

Task 3

a)

The goal of this task is to show that a parameter vector $\boldsymbol{\beta}$ which minimises the expected squared error $E[\boldsymbol{e^T e}]$ implies the orthogonality condition

$$\boldsymbol{J}(\boldsymbol{\beta})^T\big(\boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\beta})\big) = \boldsymbol{0}.$$

Let the residual vector be defined as

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\beta}).$$

Then the objective function is

$$E[\boldsymbol{e^T e}] = E\big[(\boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\beta}))^T(\boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\beta}))\big]$$

Since $\boldsymbol{y}$ is observed (or conditioning on $\boldsymbol{x}$), minimising $E[\boldsymbol{e^T e}]$ is equivalent to minimising

$$S(\boldsymbol{\beta}) = \boldsymbol{e}^T\boldsymbol{e}.$$

Substituting $\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\beta})$ yields

$$S(\boldsymbol{\beta}) = (\boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\beta}))^T(\boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\beta})).$$

Expanding this expression gives

$$S(\boldsymbol{\beta}) = \boldsymbol{y}^T\boldsymbol{y} - 2\boldsymbol{y}^T\boldsymbol{m}(\boldsymbol{\beta}) + \boldsymbol{m}(\boldsymbol{\beta})^T\boldsymbol{m}(\boldsymbol{\beta}).$$

Define the Jacobian matrix

$$\boldsymbol{J}(\boldsymbol{\beta}) = \frac{\partial \boldsymbol{m}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta^T}}.$$

Note that $\boldsymbol{y}^T\boldsymbol{y}$ does not depend on $\boldsymbol{\beta}$, and therefore its derivative with respect to $\boldsymbol{\beta}$ is zero. Differentiating $S(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ gives

$$\begin{aligned}\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} &= -2\boldsymbol{J}(\boldsymbol{\beta})^T\boldsymbol{y} + 2\boldsymbol{J}(\boldsymbol{\beta})^T\boldsymbol{m}(\boldsymbol{\beta}) \\ &= -2\boldsymbol{J}(\boldsymbol{\beta})^T\big(\boldsymbol{y} - \boldsymbol{m}(\boldsymbol{\beta})\big).\end{aligned}$$

At a minimiser $\hat{\boldsymbol{\beta}}$, the first-order condition requires

$$\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\Big|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}} = \boldsymbol{0}.$$

Hence,

$$\begin{aligned}\boldsymbol{0} &= -2\boldsymbol{J}(\hat{\boldsymbol{\beta}})^T\big(\boldsymbol{y} - \boldsymbol{m}(\hat{\boldsymbol{\beta}})\big), \\ \boldsymbol{0} &= \boldsymbol{J}(\hat{\boldsymbol{\beta}})^T\big(\boldsymbol{y} - \boldsymbol{m}(\hat{\boldsymbol{\beta}})\big).\end{aligned}$$

Minimising the expected squared error leads to the first-order equations of the nonlinear least squares problem. These equations imply that the residual vector is orthogonal to the column space of the Jacobian $\boldsymbol{J}(\boldsymbol{\beta})$, showing the required orthogonality condition.

b)

Assume linear model

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\mathbf{y} \in R^{n\times 1}$, $X \in R^{n\times p}$ and $\boldsymbol{\beta} \in R^{p\times 1}$.

NLS can be written as:

$$\mathbf{y} = \mathbf{m}(\boldsymbol{\beta}) + \boldsymbol{\varepsilon},$$

where $\mathbf{m}(\boldsymbol{\beta}) \in R^{n\times 1}$.

For OLS:

$$\mathbf{m}(\boldsymbol{\beta}) = \boldsymbol{X\beta}.$$

Jacobian is defined as:

$$J(\boldsymbol{\beta}) = \frac{\partial \mathbf{m}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^T}.$$

$$\mathbf{m}(\boldsymbol{\beta}) = \boldsymbol{X\beta}.$$

$$J(\boldsymbol{\beta}) = \boldsymbol{X}.$$

NLS-estimator minimizes sum of squares

$$J(\hat{\boldsymbol{\beta}})^T\big(\mathbf{y} - \mathbf{m}(\hat{\boldsymbol{\beta}})\big) = 0.$$

For $\mathbf{m}(\boldsymbol{\beta}) = \boldsymbol{X\beta}$ and $J(\boldsymbol{\beta}) = \boldsymbol{X}$ we get

$$\boldsymbol{X^T}\big(\mathbf{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}\big) = 0.$$

$$\boldsymbol{X^T X}\hat{\boldsymbol{\beta}} = \boldsymbol{X^T}\mathbf{y}.$$

$$\hat{\boldsymbol{\beta}}_{OLS} = \boldsymbol{(X^T X)}^{-1}\boldsymbol{X^T}\mathbf{y}.$$

Which is the solution for OLS coefficient.

c)

i.

The task is to derive the Jacobian $\boldsymbol{J(\beta)}$ mathematically.

We have:

$$m_i(\boldsymbol{\beta}) = \beta_0 \exp[(\beta_1 x_{i1} + \beta_2 x_{i2})]$$

The Jacobian is defined as:

$$\boldsymbol{J(\beta)} = \frac{\partial \boldsymbol{m(\beta)}}{\partial \boldsymbol{\beta^T}}$$

For row i, the partial derivatives becomes:

$$\begin{pmatrix} \frac{\partial m_i(\boldsymbol{\beta})}{\partial \beta_0} & \frac{\partial m_i(\boldsymbol{\beta})}{\partial \beta_1} & \frac{\partial m_i(\boldsymbol{\beta})}{\partial \beta_2} \end{pmatrix}$$

If we derivate with respect to each β we get:

$$\frac{\partial m_i(\boldsymbol{\beta})}{\partial \beta_0} = \exp(\beta_1 x_{i1} + \beta_2 x_{i2})$$

$$\frac{\partial m_i(\boldsymbol{\beta})}{\partial \beta_1} = \beta_0 x_{i1} \exp(\beta_1 x_{i1} + \beta_2 x_{i2})$$

$$\frac{\partial m_i(\boldsymbol{\beta})}{\partial \beta_2} = \beta_0 x_{i2} \exp(\beta_1 x_{i1} + \beta_2 x_{i2})$$

The Jacobian for a row i then becomes:

$$J_i(\beta) = \begin{pmatrix} \exp(\beta_1 x_{i1} + \beta_2 x_{i2}) & \beta_0 x_{i1} \exp(\beta_1 x_{i1} + \beta_2 x_{i2}) & \beta_0 x_{i2} \exp(\beta_1 x_{i1} + \beta_2 x_{i2}) \end{pmatrix}$$

The Jacobian becomes:

$$\boldsymbol{J}(\boldsymbol{\beta}) = \begin{pmatrix} \exp(\beta_1 x_{11} + \beta_2 x_{12}) & \beta_0 x_{11} \exp(\beta_1 x_{11} + \beta_2 x_{12}) & \beta_0 x_{12} \exp(\beta_1 x_{11} + \beta_2 x_{12}) \\ \exp(\beta_1 x_{21} + \beta_2 x_{22}) & \beta_0 x_{21} \exp(\beta_1 x_{21} + \beta_2 x_{22}) & \beta_0 x_{22} \exp(\beta_1 x_{21} + \beta_2 x_{22}) \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \exp(\beta_1 x_{N1} + \beta_2 x_{N2}) & \beta_0 x_{N1} \exp(\beta_1 x_{N1} + \beta_2 x_{N2}) & \beta_0 x_{N2} \exp(\beta_1 x_{N1} + \beta_2 x_{N2}) \end{pmatrix}$$

ii.

The identifiability condition is simplified to

$$E_{\beta_1}(\boldsymbol{y} \mid \boldsymbol{x}) = E_{\beta_2}(\boldsymbol{y} \mid \boldsymbol{x}) \Rightarrow \boldsymbol{\beta_1} = \boldsymbol{\beta_2}$$

Meaning that the equality of the conditional expectations results in equality of the parameters.

We have the model from the conditional expectation from the exercise:

$$E(y_i \mid x_i) = m_i(\beta) = \beta_0 \exp(\beta_1 x_{i1} + \beta_2 x_{i2}) \quad \text{for all } i, ..., n.$$

We start with the equality of the conditional expectations:

$$\beta_0^{(1)} \exp(\beta_1^{(1)} x_{i1} + \beta_2^{(1)} x_{i2}) = \beta_0^{(2)} \exp(\beta_1^{(2)} x_{i1} + \beta_2^{(2)} x_{i2}) \quad \text{for all } i, ..., n.$$

Take the log to transform into a linear expression:

$$\log(\beta_0^{(1)}) + \beta_1^{(1)} x_{i1} + \beta_2^{(1)} x_{i2} = \log(\beta_0^{(2)}) + \beta_1^{(2)} x_{i1} + \beta_2^{(2)} x_{i2}$$

Move all terms to one side and collect the terms:

$$(\log(\beta_0^{(1)}) - \log(\beta_0^{(2)})) + (\beta_1^{(1)} x_{i1} - \beta_1^{(2)} x_{i1}) + (\beta_2^{(1)} x_{i2} - \beta_2^{(2)} x_{i2}) = 0$$

$$(\log(\beta_0^{(1)}) - \log(\beta_0^{(2)})) + x_{i1}(\beta_1^{(1)} - \beta_1^{(2)}) + x_{i2}(\beta_2^{(1)} - \beta_2^{(2)}) = 0$$

We have from the exercise that we should assume that the matrix with rows $(1, x_{i1}, x_{i2})$ has full column rank.

We set

$$a = (\log(\beta_0^{(1)}) - \log(\beta_0^{(2)}))$$

$$b = (\beta_1^{(1)} - \beta_1^{(2)})$$

$$c = (\beta_2^{(1)} - \beta_2^{(2)})$$

We then have:

$$a + bx_{i1} + c_{xi2} = \begin{pmatrix} 1 & x_{i1} & x_{i2} \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0$$

Which we could write as:

$$\begin{pmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ . & . & . \\ . & . & . \\ . & . & . \\ 1 & x_{N1} & x_{N2} \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ . \\ . \\ . \\ 0 \end{pmatrix}$$

Since the exercise said that we should assume that the matrix with rows $(1, x_{i1}, x_{i2})$ has full column rank the only solution for $a + bx_{i1} + c_{xi2} = 0$ for all i is:

$$a = b = c = 0$$

Which then must mean that

$$a = (\log(\beta_0^{(1)}) - \log(\beta_0^{(2)})) = \log(\frac{\beta_0^{(1)}}{\beta_0^{(2)}}) = 0$$

$$b = (\beta_1^{(1)} - \beta_1^{(2)}) = 0$$

$$c = (\beta_2^{(1)} - \beta_2^{(2)}) = 0$$

Thus, we can see that:

$$\beta_0^{(1)} = \beta_0^{(2)}$$

$$\beta_1^{(1)} = \beta_1^{(2)}$$

$$\beta_2^{(1)} = \beta_2^{(2)}$$

We have therefore proven that the equality of the conditional expectations results in equality of the parameters

$$E_{\beta_1}(\boldsymbol{y} \mid \boldsymbol{x}) = E_{\beta_2}(\boldsymbol{y} \mid \boldsymbol{x}) \Rightarrow \boldsymbol{\beta_1} = \boldsymbol{\beta_2}$$

Meaning that the model **is identifiable** on the parameter space where $\{\beta_0 \neq 0\}$.

If $\beta_0 = 0$ the model will collapse and β_1 and β_2 will have no effect on the model. This means that different values for these parameters would yield the same conditional mean, meaning the model is **not identifiable** for this special case where $\beta_0 = 0$.